

Gergely Szabo

WEB BACKEND DEVELOPER · BLOCKCHAIN DEVELOPER

Gyor, Hungary

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Professional Summary

I am a Software Engineer mainly specialised in blockchain dApp and web backend development.

Work Experience

Haskell/Cardano Blockchain Consultant

MLabs, Gyor, Hungary (remote)

FULL TIME

Aug 2021 - present

- Led design and implementation of an off-chain tool in Haskell for building, batching and submitting transactions, handling millions of dollars worth of tokens safely.
- Developed a Rust-based off-chain toolkit including a transaction builder, chain-follower and indexer, improving development workflows and computational resource requirements. (Rust, PostgreSQL)
- Engineered Rust code generation for LambdaBuffers schema language compiler, facilitating type-safe integration in a polyglot environment.
- Worked as technical lead and project manager delivering Plutus smart contracts and CLI tools for IOG's partnerchain project, following agile methodologies with weekly sprint meetings and story points.
- Designed and implemented several Cardano dApps (Plutus, Plutarch, Servant, REST, gRPC)
- Co-designed and implemented an extension to the Agora on-chain voting platform, integrating with L1 dRep voting.
- Audited dApps finding security vulnerabilities and suggesting solutions.
- Implemented and scaled OpenFGA-based authorization across backend workflows.
- Refactored Haskell backend modules toward ports-and-adapters architecture.

Full Stack Web Engineer

Kakekomu Inc., Tokyo, Japan (remote)

PART TIME UNTIL MAR 2019, THEN FULL TIME

Jan 2018 - Jul 2021

- Designed and implemented a Q&A service backend/frontend using TypeScript, Koa.js, MySQL and Next.js, deployed to AWS ECS.
- Configured CI/CD flows for several services.
- Implemented the front end of a lawyer search platform in Next.js.
- Created an internal admin tool using Elm and Ruby on Rails.
- Built an internal UI library using React, published to a private GitHub npm registry.
- Worked in a remote team using agile methodologies.

Frontend Web Engineer

Yahoo Japan, Tokyo, Japan

FULL TIME

Jan 2018 - Mar 2019

- Implemented new features on an advertisement management tool using React.
- Configured CI flow for the advertisement management tool using Screwdriver.
- Worked in both waterfall and agile workflows during the company's agile transition.

Junior Software Engineer

Happiness Technology, Tokyo, Japan

FULL TIME

Apr 2017 - Dec 2017

- Implemented new features on an inventory management system (Java, Oracle PL/SQL, Flash, JavaScript).

Education

Kyoto University of Education

JAPANESE LANGUAGE AND CULTURAL STUDIES EXCHANGE STUDENT

Kyoto, Japan

2013-2014

A Tan Kapuja Buddhista College

BA IN JAPANESE LANGUAGE AND CULTURE

Budapest, Hungary

2011 - 2015

Projects

LambdaBuffers

SCHEMA LANGUAGE AND CODE GENERATOR FOR POLYGLOT PROJECTS

- <https://github.com/mlabs-haskell/lambda-buffers>

Haskell / Rust / Nix

Plutus Ledger API

PLUTUS TYPES AND USEFUL TOOLS FOR CARDANO DAPP DEVELOPMENT

- <https://github.com/mlabs-haskell/plutus-ledger-api-rust>

Rust

Tx Village

RUST BASED TOOLKIT FOR CARDANO TRANSACTION BUILDING, VERIFICATION AND CHAIN-INDEXING

- <https://github.com/mlabs-haskell/tx-village>

Rust

Bot Plutus Interface

TRANSACTION BUILDER FRAMEWORK BASED ON PLUTUS-APPS CONTRACT MONAD INTERFACE

- <https://github.com/mlabs-haskell/bot-plutus-interface>

Haskell

Plutip

CARDANO LOCAL TEST NETWORK EXECUTER AND TEST FRAMEWORK

- <https://github.com/mlabs-haskell/plutip>

Haskell

Cardano Devnet Flake

CARDANO LOCAL TEST NETWORK EXECUTER BASED ON PROCESS-COMPOSE

- <https://github.com/mlabs-haskell/cardano-devnet-flake>

Nix

Flake-lang

NIX TOOLS POWERING POLYGLOT MONO-REPOSITORIES

- <https://github.com/mlabs-haskell/flake-lang.nix>

Nix

Skills

Languages

Haskell, Rust, TypeScript, Nix, Elm

Backend

Servant, Axum, Express, gRPC, REST APIs, PostgreSQL

Blockchain

Cardano, Plutus, Plutarch, Solidity

Frontend

React, Elm, Next.js

DevOps

Linux, NixOS, Docker, AWS, GCP, Terraform, Hercules CI, Cachix

Testing

property-based testing, designing easily testable abstractions

Agentinc development

Claude Code, Codex, open-code

Languages

Fluent

Hungarian (native), English (TOEIC 985), Japanese (JLPT N1)

Currently learning French

Interests

road cycling, chess, programming, bass guitar (jazz/fusion), language learning